

### PROBABILITY 3: ASSIGNMENT 4

In what follows,  $\mathcal{P}(\mathbb{R})$  denote the space of Borel probability measures on  $\mathbb{R}$ . By  $\mu_n \xrightarrow{d} \mu$ , we will mean *convergence in distribution*.

**Problem 1.** Define

$$\mu_n = \frac{2}{n(n+1)} \sum_{k=1}^n k \delta_{k/n}.$$

Then show that  $\mu_n \xrightarrow{d} \mu$  as  $n \rightarrow \infty$  for some  $\mu \in \mathcal{P}(\mathbb{R})$  (which you must identify explicitly).

**Problem 2.** Suppose  $\mu_n, \mu \in \mathcal{P}(\mathbb{R})$  and that  $F_\mu$  is continuous. If  $\mu_n \xrightarrow{d} \mu$ , show that  $F_{\mu_n}(t) - F_\mu(t) \rightarrow 0$  uniformly over  $t \in \mathbb{R}$ .

**Problem 3.** Suppose  $\mu_n, \mu$  are discrete probability measures supported on  $\mathbb{Z}$  having probability mass functions  $(p_n(k))_{k \in \mathbb{Z}}$  and  $(p(k))_{k \in \mathbb{Z}}$ . Show that  $\mu_n \xrightarrow{d} \mu$  if and only if  $p_n(k) \rightarrow p(k)$  for each  $k \in \mathbb{Z}$ .

**Problem 4.** True or False: Let  $\mu_n, \mu \in \mathcal{P}(\mathbb{R})$  and assume that  $F_{\mu_n}(x) \rightarrow F_\mu(x)$  for all  $x \in D$ , a countable dense subset of  $\mathbb{R}$ , then  $\mu_n \xrightarrow{d} \mu$ .

**Problem 5.** Give examples of measures  $\mu_n, \mu$  such that  $\mu_n \xrightarrow{d} \mu$ , but  $F_{\mu_n}(x) \not\rightarrow F_\mu(x)$  for some  $x \in \mathbb{R}$ .